Characterizing the Dark Signal

Six tests, in order, and the two decisions they led to

Colorimeter project — pad analysis pipeline

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Abstract

Every measurement this instrument makes starts by subtracting a "dark" exposure — a frame read out with no light reaching the sensor — from the corresponding lit frame, on the reasoning that whatever the sensor reports with no light present is an offset to be removed, not signal. Until now that subtraction has never been examined on its own: we always looked at it as part of the larger reflectance calculation. This document works through six increasingly specific questions about the dark signal, each posed in plain language, answered with an explicit and elementary statistical method, and illustrated with a figure produced by `desktop/padanalysis/analysis_2.py`. The questions build on one another: is a single dark frame just noise, or does it have structure; if it has structure, is that structure the same pixel to pixel and stable over time; does it drift; does it depend on how long the exposure was; and, putting all of that together, what is actually the best thing to subtract from a future lit frame. Two practical decisions follow directly from the answers, and both are stated at the end: subtract a fresh dark frame's own average level rather than its raw pixel-by-pixel values, except at a small, explicitly identified set of pixels that behave differently; and replace the present ladder of many exposure times with a pair of "before" and "after" dark batches, since the dark level turns out not to depend on exposure duration by an amount worth chasing.

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1 Introduction and scope

1.1 Why this document exists

The photometric chain used elsewhere in this project (see `tex/R_inf_guide.tex`) begins with `sig = lit - dark`: every downstream quantity, from the raw ratio `R` through `A` and `K/S`, inherits whatever error is in that one subtraction. Nobody had asked, with real data and a real method, how good that subtraction actually is, or whether it could be made better cheaply. This document is the record of asking. It follows the "show the method, not just the answer" habit already established for this project's analysis notes: each section states why a question mattered, what test was chosen to answer it, how that test was implemented, and what the result actually was — including where a result surprised us or needed a second look.

1.2 What data this is based on

Every number and figure below comes from nine archives, `C1017` through `C1025`, read through channel `B` (one of the four Bayer layers; see §2) of the measurement sweep. Two of the nine (`C1017`, `C1018`) are "exposure-ladder" captures that took a dark frame at 22 different exposure times; the remaining seven took a dark frame at 8 exposure times each. Pooled, this gives 100 stored dark frames, each 130×183 pixels (23,790 pixels) in the bound-crop layer grid this project works in. This is a convenience sample, not a designed experiment — it is what happened to be captured for an unrelated purpose (the round-dot dye series) — and every conclusion below should be read with that in mind. Two limitations are worth stating up front rather than discovering later: everything here is channel `B` only, and every dark frame used is an "after" frame (see §10); the older "before/after" pairing this document ultimately recommends reinstating is not itself present in this sample.

1.3 Where the code lives

Every function named below is in `desktop/padanalysis/analysis_2.py`. It is read-only against the archives and writes nothing but figures, so every result here can be reproduced by calling the named function with the same sample list.

2 Notation and terms

Some of this is camera terminology, some is statistics terminology. Both are defined here once, and used consistently afterward.

2.1 The data

DN "Digital number," the raw unit a pixel is read out in. This sensor's analog-to-digital converter is 10-bit, so a single raw reading is an integer from 0 to 1023.

Bayer layer A color camera sensor covers its pixels with a repeating 2×2 pattern of color filters: one red, one blue, and two green (traditionally called `Gr`, the green filter sharing a row with red, and `Gb`, the green filter sharing a row with blue). This project treats the four as four separate images, called *layers*, each at half the sensor's linear resolution. §2 onward writes ch $\in \{R, Gr, Gb, B\}$ for which layer, and every result in this document uses ch $= B$.

Bound-crop layer grid The small rectangular region of one layer that actually contains the test strip, in

that layer's own row/column coordinates. For this sample set it is 130 rows by 183 columns, i.e. $N_{\text{px}} = 23{,}790$ pixels.

Exposure How long the sensor integrates light (or, for a dark frame, how long it integrates nothing) before being read out, in microseconds (μs). One archive stores one dark frame per distinct exposure time used anywhere in its lit sweep; ten LEDs sharing the same exposure time share that one dark frame (`probe.py`, `Capture._dark_link`).

Archive / sample
One capture, e.g. `C1017`, stored as one HDF5 file. Everything in this document uses the "measurement" section of the archive (as opposed to the shorter "pre-image" section; see `probe.py`'s module docstring).

Pixel index i
A single location in the bound-crop layer grid. Where a formula sums or averages "over i," it means over all N_{px} pixels of that grid.

Frame index k
One particular stored dark frame, $k = 1, \ldots, n$, where n is however many dark frames a given calculation pools together (e.g. $n = 100$ for the nine-archive pool used throughout this document). Do not confuse n (a count of frames) with N_{px} (a count of pixels) — both appear in the same formulas below.

$d_i(k)$ The dark-subtracted-nothing, i.e. raw, reading of pixel i in frame k, in DN. This is what `probe.Capture.darklayer` returns.

2.2 The statistics

Each of these is also introduced again, briefly, the first time it is used; this is a single place to look them back up.

Mean, variance, standard deviation
For a list of M numbers $x_1, \ldots, x_M$: the mean is $\bar{x} = \frac{1}{M}\sum_j x_j$; the variance is $\mathrm{Var}(x) = \frac{1}{M-1}\sum_j (x_j - \bar{x})^2$ (dividing by $M-1$ rather than M is the small, standard correction that makes this an unbiased estimate of the true population variance when $\bar{x}$ itself was estimated from the same data); the standard deviation σ is the square root of the variance, in the same units as the data.

Pearson correlation coefficient, r
For two paired lists a_j, b_j, $r = \mathrm{cov}(a,b)/(\sigma_a \sigma_b)$, where $\mathrm{cov}(a,b) = \frac{1}{M-1}\sum_j (a_j - \bar{a})(b_j - \bar{b})$. r ranges from -1 to 1; r^2 is the fraction of one list's variance that is shared with (linearly predictable from) the other.

Empirical cumulative distribution function (CDF)
For N sorted values $x_{(1)} \le x_{(2)} \le \cdots \le x_{(N)}$, the empirical CDF at $x_{(i)}$ is $F_{\text{emp}}(x_{(i)}) = (i - \frac{1}{2})/N$: the fraction of the data at or below that value. It is the data's own answer to "what fraction of values are this small or smaller."

Normal (Gaussian) distribution and erf
The familiar bell curve with mean μ and standard deviation σ. Its CDF has a closed form written in terms of the error function, $\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}\,dt$: $F_{\text{normal}}(x) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$.

Permutation test

A way to ask "is this pattern more than I would see from shuffled, meaningless data," without assuming the data are normally distributed. Compute the statistic of interest on the real data; then shuffle the labels (e.g. which exposure or which time each value is attached to) many times, recomputing the same statistic each time, to build up a distribution of what the statistic looks like when there is, by construction, no real relationship. The fraction of shuffled trials whose statistic is at least as extreme as the real one is the (Monte Carlo) p-value.

One-way ANOVA and the F-statistic

A way to ask whether several groups of numbers have different means, given only how spread out each group is internally. Defined and derived where first used, §8.

Statistical vs. practical significance

A result can be statistically detectable (unlikely to be pure chance) while still being too small to matter for any practical purpose. Both a p-value and an effect size (how big the difference actually is, in DN) are reported together below for exactly this reason.

3 A first look at one dark frame

Why we asked this. Before any statistics, the most basic question is simply: what does a dark frame look like? Is it visually uniform, or does it have obvious structure? What is a typical value, and how much do pixels vary around it?

What we did. `dark_analysis1(samp_no, exposure_ms, ch)` reads one dark frame through `probe.Capture.darklayer` and draws it two ways side by side: as an image, and as the same pixel values sorted from darkest to brightest ($d_{(1)} \le d_{(2)} \le \cdots \le d_{(N_{\mathrm{px}})}$). The sorted view turns 23,790 numbers into one line: a flat middle with short ramps at each end is ordinary noise sitting on a uniform offset; a long slope, a step, or an isolated jump at either end is something structural — a hot pixel at the top, a cold or dead one at the bottom.

What we found. Figure 1 shows one dark frame from `C1017` at a 50.973 ms exposure. Two things stand out, and both matter for everything that follows. First, the frame looks like ordinary speckle: no visible large-scale pattern, no obviously distinct pixels beyond a handful of very faint bright specks. Second, and easy to miss: the sorted curve on the right is a *staircase* with only five steps, at 62, 63, 64, 65, and 66 DN. A single raw dark reading is a 10-bit integer, and with a mean near 64 DN and a spread of well under one DN, the entire 23,790-pixel frame only occupies about five integer values. This is not a defect; it is simply the resolution of one raw reading, and it becomes important again in §5, where it explains a pattern that could otherwise be mistaken for something more interesting.

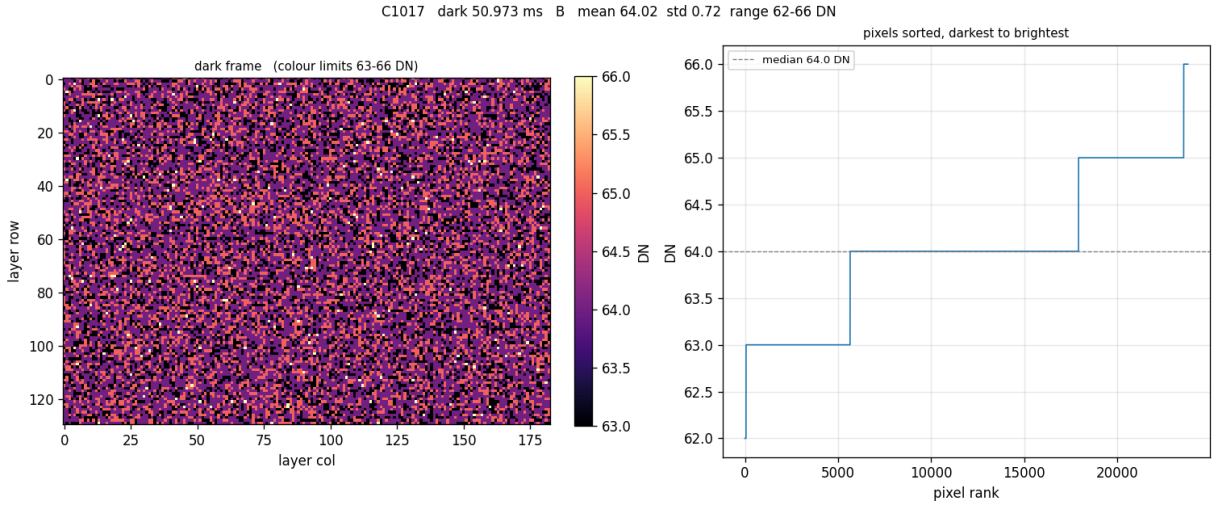


Figure 1: One raw dark frame, `C1017`, exposure 50.973 ms, channel `B`. Left: the frame as an image. Right: the same 23,790 pixel values sorted from darkest to brightest — note the staircase, a direct consequence of the sensor's 10-bit integer readout.

4 Is there a real pixel-to-pixel pattern, or is it all noise?

Why we asked this. One frame cannot answer this. A pixel that reads a little high in one frame might just be having a noisy moment; only by looking at the same pixel across many frames can "consistently a little high" be told apart from "high just this once."

The model. We adopt the simplest structure that lets that question be asked at all: for pixel i in frame k,

$$d_i(k) \;=\; \mu(k) \;+\; p_i \;+\; \varepsilon_i(k), \tag{1}$$

where

- $\mu(k)$ is frame k's own *spatial mean* — the average of $d_i(k)$ over all pixels i in that one frame. It is a property of the frame, not of any pixel, and it is the term that can drift with time or with exposure (§7, §8).
- p_i is pixel i's own fixed, time-invariant offset from the frame mean — the "fixed pattern." If p_i is a real, reproducible property of the pixel, it should look the same no matter which frames are used to estimate it; §6 tests exactly that.
- $\varepsilon_i(k)$ is whatever is left over: fresh, independent noise, one new random draw for this pixel in this frame, uncorrelated with p_i and with ε_i in any other frame. (Earlier work on this project established that $\varepsilon_i(k)$ is also spatially independent — knowing one pixel's noise says nothing about its neighbor's — so nothing here needs to borrow strength across neighboring pixels.)

This is nothing more than "every reading is a frame-level average, plus a personal bias for this pixel, plus leftover noise" written as an equation. Everything from here on is about estimating the three pieces on the right separately, because they behave differently and, as §9 shows, deserve different treatment.

What we did. `dark_analysis2(samps, ch)` stacks every stored dark frame for the given samples and channel into one array, indexed by pixel and frame, and computes two per-pixel summaries across the n frames: the mean $\bar{d}_i = \frac{1}{n}\sum_k d_i(k)$ and the standard deviation s_i of $d_i(k)$

over k. Sorting pixels by $\bar{d}_i$ and plotting both quantities against that sorted order is a direct visual test of Equation 1: if p_i were exactly zero for every pixel, the sorted-mean curve would be flat (up to noise); if $\varepsilon_i(k)$ did not depend on the pixel's own level, s_i would also be flat across the sort.

What we found. Figure 2 shows the result pooling all nine archives, $n = 100$ frames. The top-left panel is the mean frame itself; on its own percentile-based color scale it shows the same faint speckle as Figure 1, now averaged down to a much smaller range (63.8–64.3 DN). The top-right panel is the telling one: the sorted mean is very nearly flat across the great majority of pixels, climbing gently, but rises sharply at the extreme right — a small number of pixels sit clearly above the rest, both in level and, in the bottom panels, in their own standard deviation as well. Both bottom panels show the same thing at different zoom levels: the noise level s_i is essentially constant (around 0.72–0.73 DN) across the bulk of the population, and only rises for that same handful of pixels at the far right. This is the first evidence for a genuine, small population of "hot" pixels, distinct from the ordinary noise floor — a claim that §6 tests directly, rather than by eye.

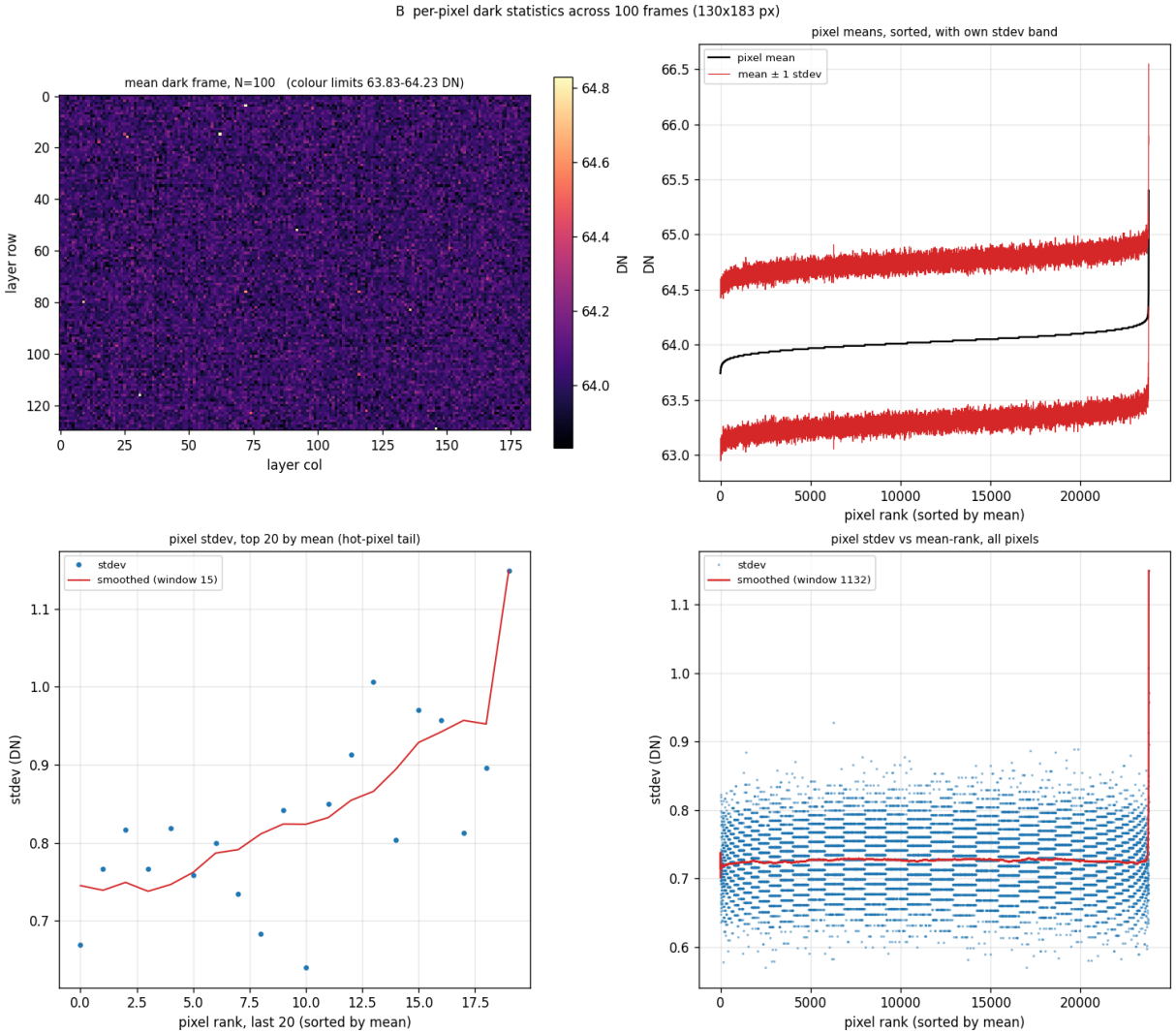


Figure 2: Per-pixel dark statistics across all $n = 100$ pooled frames, channel B. Top left: the mean frame. Top right: every pixel's mean ± one of its own standard deviations, sorted by mean. Bottom: the same per-pixel standard deviation, zoomed to the top 20 pixels by mean (left) and shown for the whole population with two smoothing windows (right).

5 Is the pixel-to-pixel pattern actually Gaussian noise?

Why we asked this. The sorted mean curve in Figure 2 has the shape one would expect if $\bar{d}_i$ were normally distributed across pixels — S-shaped when plotted this way is what a sorted sample of normal draws looks like. But "looks like" is an eyeball judgment, and the estimator built in §9 explicitly assumes p_i is normally distributed across the population of pixels. That assumption is worth checking on its own terms, quantitatively, rather than taken on faith.

What we did. For a sorted sample $x_{(1)} \le \cdots \le x_{(N)}$ (here, either the per-pixel means or the per-pixel standard deviations from §4, with $N = N_{\mathrm{px}} = 23{,}790$), we compare the empirical CDF defined in §2 directly against the CDF of the best-fitting normal distribution, using that same sample's own mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$:

$$F_{\mathrm{emp}}(x_{(i)}) = \frac{i - \frac{1}{2}}{N}, \qquad F_{\mathrm{normal}}(x) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{x - \hat{\mu}}{\hat{\sigma}\sqrt{2}}\right)\right].$$

Plotting both curves together is a direct visual test. Plotting their difference, $F_{\mathrm{emp}} - F_{\mathrm{normal}}$, against x makes the same test quantitative: its largest magnitude anywhere is a Kolmogorov–Smirnov-style distance, and, importantly, *where* it occurs says something about *what kind* of deviation is present. This is implemented in `dark_cdf_check(samps, ch, quantity)`, with `quantity = "mean"` or `"std"` selecting which per-pixel statistic is tested.

A caveat, stated rather than skipped. $\hat{\mu}$ and $\hat{\sigma}$ are estimated from the very sample being tested. A more careful procedure (a Lilliefors test) accounts for the extra flexibility this gives the fitted curve and uses a smaller, corrected critical value than the textbook Kolmogorov threshold of $1.36/\sqrt{N}$. We do not compute that correction here; the number below should be read as a descriptive distance, not a calibrated p-value.

What we found — and a correction to how we first read it. Figure 3 shows the check on the per-pixel mean. The two CDFs agree closely almost everywhere; the maximum deviation is small, 0.035, at $x = 64.06$ DN. Our first pass at reading this kind of plot assumed a deviation like this would sit in one tail and would mark the hot-pixel population already seen in Figure 2. Looking at the actual right-hand panel corrects that assumption: the deviation is not a single bump in a tail but a rapid, symmetric zig-zag centered on the *bulk* of the distribution, near its mean, exactly where the data are most dense. That shape has a specific, mundane explanation rather than a statistical one: $\bar{d}_i$ is an average of n ten-bit integers, so it can only take values that are multiples of $1/n$; with $\hat{\sigma}$ of only 0.078 DN (mean case) or 0.043 DN (standard-deviation case), the whole distribution spans only a few dozen of those discrete levels. A genuinely continuous normal distribution compared against data that can only land on a comb of discrete values will always show exactly this kind of oscillating deviation, largest where the data are densest, no matter how large the sample gets. The hot-pixel evidence stands — Figures 2 and 4 (next section) show it directly — but this particular figure is not where to look for it; it mainly demonstrates that quantization, not a hot-pixel tail, is the dominant departure from normality at the bulk of the distribution.

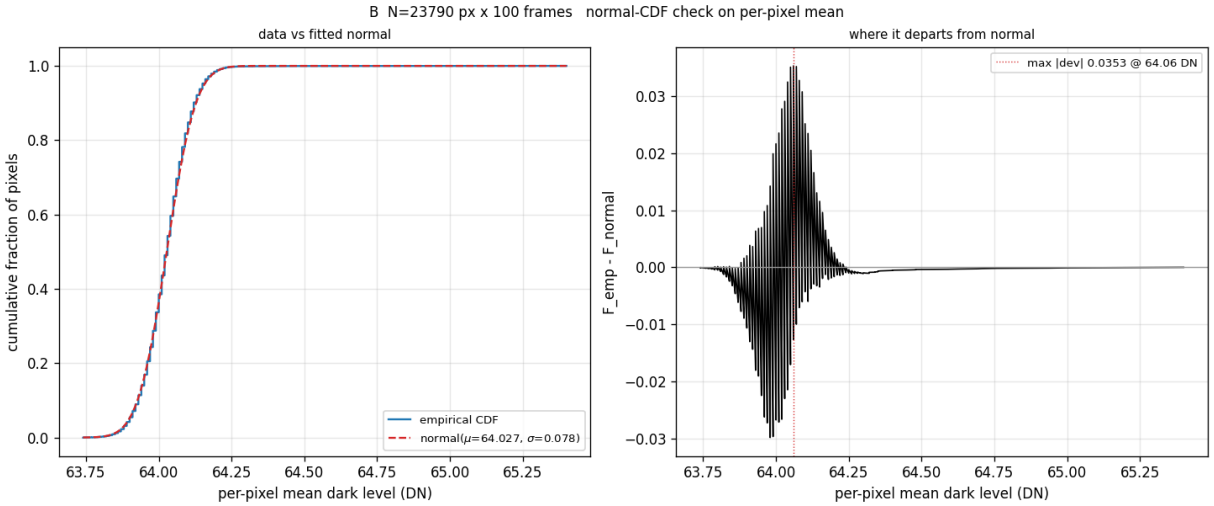


Figure 3: Normal-CDF check on the per-pixel mean dark level, $n = 100$ frames, $N_{\text{px}} = 23{,}790$ pixels. Left: empirical CDF against the best-fit normal. Right: their difference. The rapid oscillation, centered on the bulk rather than a tail, is a signature of the data's integer quantization (§3), not of a hot-pixel population.

6 Is the pattern stable over time, or just luck in this sample?

Why we asked this. A pixel could look "hot" in Figure 2 purely by chance — with 100 frames and 23,790 pixels, some pixels will land at the extreme of the distribution just from ordinary noise, the same way someone in a large enough crowd will flip ten coins and get all heads. The only way to tell a real, reproducible pixel property from a chance extreme is to check whether it shows up again independently.

What we did: split-half reliability. This is a standard trick from measurement theory, elementary once stated: split the available history into two independent halves (here, by time — the first $n/2$ frames and the last $n/2$), estimate each pixel's offset separately in each half, and see how well the two independent estimates agree. First remove the frame-level term using each frame's own spatial mean, $r_i(k) = d_i(k) - \mu(k)$; a real, reproducible p_i should agree between the two halves; noise should not. Concretely, with a_i the average of $r_i(k)$ over the first half and b_i the average over the second half, the Pearson correlation $r = \text{corr}(a, b)$ (§2) across all pixels is the fraction of each half's variance that the two halves share — i.e., the fraction that is a real, stable property of the pixel rather than each half's own independent noise. This is implemented in `dark_stability_check(samps, ch)`.

A cross-check built in for free. This same quantity — the fraction of variance that is real signal versus noise — is exactly the shrinkage weight B derived independently in §9, evaluated with n set to half the full history instead of all of it. The two methods use completely different arithmetic (one a correlation between two halves, the other a variance decomposition on the full history) and should agree once that factor of two in n is accounted for; where they do, each is a check on the other.

What we found. Figure 4 shows both halves plotted against each other, and both against the full-history estimate. The overall correlation is $r = 0.072$ — deliberately small, and correctly so: §9 independently estimates that only about 13% of an individual pixel's apparent full-history offset is real signal rather than sampling noise (the weight $B = 0.131$ at $n = 100$); at half that history the corresponding figure works out to about $B_{\text{half}} \approx 0.070$, matching the measured

r closely. A low r here is not evidence that nothing is reproducible — it is exactly what a population with a small number of genuinely offset pixels sitting in a much larger population of pixels with almost no fixed offset should produce. That structure is visible directly in the left panel of Figure 4: the great bulk of pixels forms an uncorrelated cloud centered near the origin (their apparent offsets are mostly noise, as expected), while a small number of points — most strikingly one pixel near $(1.4, 1.4)$ DN — sit far from the bulk and close to the diagonal, meaning both independent halves of the history agree closely on a large offset for that pixel. That is the signature of a real, reproducible hot pixel, and the same handful appear at the extreme right of Figure 2.

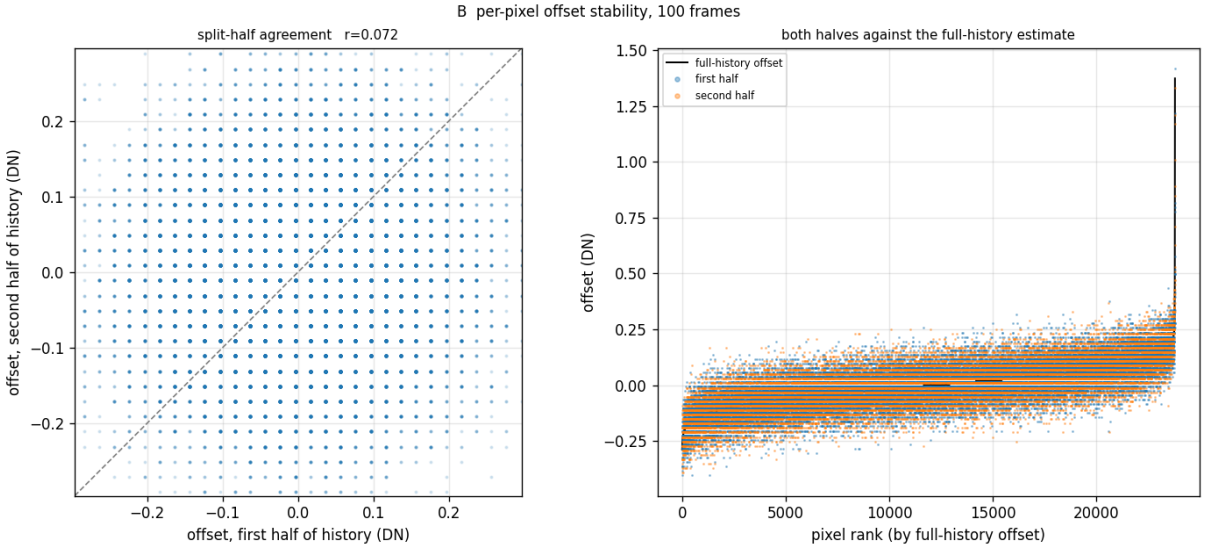


Figure 4: Split-half reliability of the per-pixel offset, $n = 100$ frames split 50/50 by time. Left: first-half offset vs. second-half offset, per pixel; the dashed line is the identity. Right: both halves and the full-history estimate, all sorted by the full-history offset.

7 Does the dark level drift with time?

Why we asked this. Three timescales are relevant to how a dark correction is actually applied in practice: seconds, within one archive's own batch of dark frames; minutes, between a dark batch and the lit frames it corrects, within the same archive; and anywhere from minutes to days, between different archives. If $\mu(k)$ drifts meaningfully on any of these scales, a dark frame taken at the wrong time is a source of error even if every pixel's p_i is estimated perfectly.

What we did. For each timescale, we regress the frame-level mean $\mu(k)$ against elapsed time $t(k)$ (seconds within one archive for the short scale, real wall-clock hours across archives for the long scale) by ordinary least squares. With time centered, $t'(k) = t(k) - \bar{t}$, the least-squares slope has the simple closed form

$$\text{slope} = \frac{\sum_k t'(k)\,(\mu(k) - \bar{\mu})}{\sum_k t'(k)^2} = \frac{\operatorname{cov}(t, \mu)}{\operatorname{Var}(t)},$$

i.e. the covariance between time and level, divided by the variance of time alone. Significance is judged by a permutation test (§2): shuffle the frame-level means against the fixed time values 2000 times, recompute the slope each time, and see where the real slope falls among the shuffled ones. This sidesteps having to assume the frame-mean residuals are themselves normally distributed, which is a separate question from the one §5 answers (that one concerns the per-pixel offset, not the frame-to-frame level). `dark_temporal_check(samps, ch)` implements both scales.

The middle timescale — dark to lit, minutes, within one archive — could not be tested on this sample. It requires two dark batches bracketing the same lit sequence, and every archive here stores only one ("after") batch. This is exactly the gap that Decision 2 (§10) closes going forward.

The long timescale needs one more precaution. A ladder archive (`C1017`, `C1018`) stores dark frames at 22 exposure times; a single-rung archive stores only 8. Averaging each archive's *entire* set of dark frames to get one "archive mean" would silently compare averages built from different mixes of exposures, confounding any real drift with a difference in composition. We instead restrict the long-scale comparison to the 8 exposure times common to every one of the nine archives, so each archive's point in that comparison is built the same way.

What we found. Figure 5 shows both scales. Within `C1017` (left), the fitted slope is +0.88 DN per 1000 s, with a permutation $p = 0.165$ — not distinguishable from no trend at all across an 11.5-second dark batch. Between the nine archives (right), spanning about 48 hours, the fitted slope is +0.0001 DN per hour, $p = 0.333$ — again consistent with no real drift once the comparison is restricted to matched exposures. The right-hand panel does show a visible sawtooth: the level jumps upward between `C1018` and `C1019`, and again between `C1022` and `C1023`, decaying gently downward in between. Both jumps are small — roughly 0.015–0.02 DN, about half of σ_p (§9) and two orders of magnitude below the single-frame noise floor σ_ε — and with only three apparent groupings in nine archives, there are far too few repeats to say whether this is a real, repeatable effect (for instance, a brief warm-up each time a capture session restarts) or simply what a handful of points looks like under ordinary sampling variation. We record it as an open question worth watching, not as a finding.

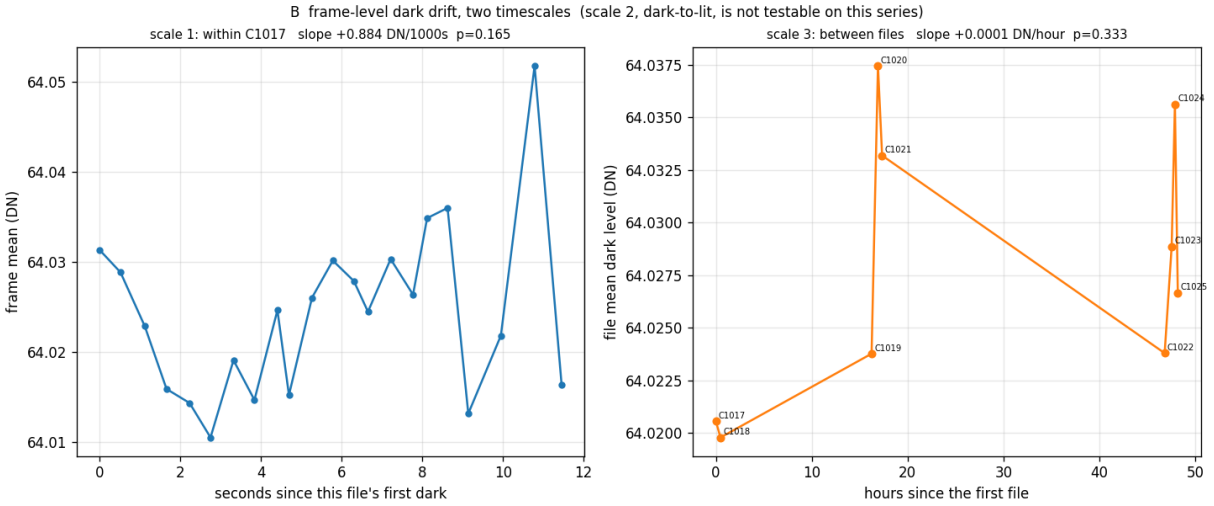


Figure 5: Frame-level dark drift. Left: within one archive (`C1017`), seconds. Right: between the nine archives, hours, restricted to the 8 exposure times common to all of them.

8 Does the dark level, or its noise, depend on exposure duration?

Why we asked this. This question has a direct, practical payoff: if the dark level (or its noise) depends meaningfully on how long the exposure was, a whole ladder of dark exposures is needed to correct every lit exposure correctly. If it does not, a single dark exposure — or a low one and a high one — is enough, freeing up capture time and storage.

A confound, faced directly. Within any one archive, the dark frames are captured in strictly increasing exposure order, so *elapsed time since the batch started* and *exposure duration* are the same variable, relabeled — no test run on one archive's own sequence can tell which of the two it is responding to. They can only be told apart where the *pairing* between exposure time and elapsed time differs across archives, and it does: a ladder archive visits 22 exposures over about 11.5 seconds; a single-rung archive visits 8 of the same exposure values over a much shorter span, at different relative positions in its own batch. Comparing many archives at the *same* exposure time, each having reached that exposure time at a different point in its own batch, is the natural comparison this data set offers. It is not a perfect one: it separates exposure duration from absolute, cross-session time, but it does not fully separate exposure duration from "elapsed time since this particular archive's own dark batch began," since every batch, regardless of how many exposures it holds, proceeds shortest-exposure-first. This limitation is stated here rather than glossed over, and the conclusion below is phrased with it in mind.

What we did: one-way ANOVA with a permutation null. Treat exposure time as a factor with $G = 8$ levels (the common exposure times), each represented by one frame from each of the 9 archives (so $G = 8$ groups of 9 frames, 72 frames total after restricting to the common exposures). "Analysis of variance" asks whether the differences *between* the group means are large compared to the scatter *within* each group. Writing $y_{g,j}$ for the j-th frame's statistic (mean or standard deviation) in exposure group g, $\bar{y}_g$ for that group's own mean, and $\bar{y}$ for the overall mean:

$$\mathrm{SS}_{\mathrm{between}} = \sum_g n_g(\bar{y}_g - \bar{y})^2, \qquad \mathrm{SS}_{\mathrm{within}} = \sum_g \sum_j (y_{g,j} - \bar{y}_g)^2,$$

with n_g the number of frames in group g (here, 9 for every group). Dividing each sum of squares by its degrees of freedom ($df_{\mathrm{between}} = G - 1$, $df_{\mathrm{within}} = \sum_g n_g - G$) gives the two average squared spreads being compared, and their ratio is the F-statistic:

$$F = \frac{\mathrm{SS}_{\mathrm{between}}/df_{\mathrm{between}}}{\mathrm{SS}_{\mathrm{within}}/df_{\mathrm{within}}}.$$

F near 1 means the between-group differences look like the same kind of scatter already present within each group — no real group effect; F much larger than 1 means the groups differ by more than their own internal scatter would predict. Because the within-group counts here are small (9 frames per group), we judge F by a permutation test rather than the textbook F-distribution: shuffle the exposure-time labels across all 72 frames (keeping the same group sizes), recompute F, 2000 times, and see where the real F falls. This is implemented as `dark_exposure_check(samps, ch, quantity)`, run once for the per-frame mean (dark current, i.e. whether charge accumulates with integration time) and once for the per-frame standard deviation (whether the noise itself, as opposed to the level, depends on exposure).

What we found. For the level (Figure 6): $F = 2.48$, permutation $p = 0.026$ — a real, statistically detectable effect, not attributable to chance at the usual 5% threshold. But the *size* of that effect is the more important number for the original question: the largest difference between any two exposure groups' means is only 0.0225 DN, smaller than σ_p itself (§9) and roughly 30 times smaller than a single frame's own noise, $\sigma_\varepsilon = 0.731$ DN. This is a clean illustration of the distinction drawn in §2 between statistical and practical significance: the effect is real enough to detect with 72 frames, and yet far too small to be worth correcting for in a subtraction that is already carrying three-tenths of a DN of irreducible frame-to-frame noise. Given the acknowledged residual confound with elapsed-time-since-batch-start, we do not claim this small effect is purely an exposure-duration effect in the strict sense — only that whatever it is, it is too small to matter operationally.

For the noise (Figure 7): $F = 0.56$, permutation $p = 0.790$ — no detectable effect at all; the per-frame standard deviation does not depend on exposure duration in this data.

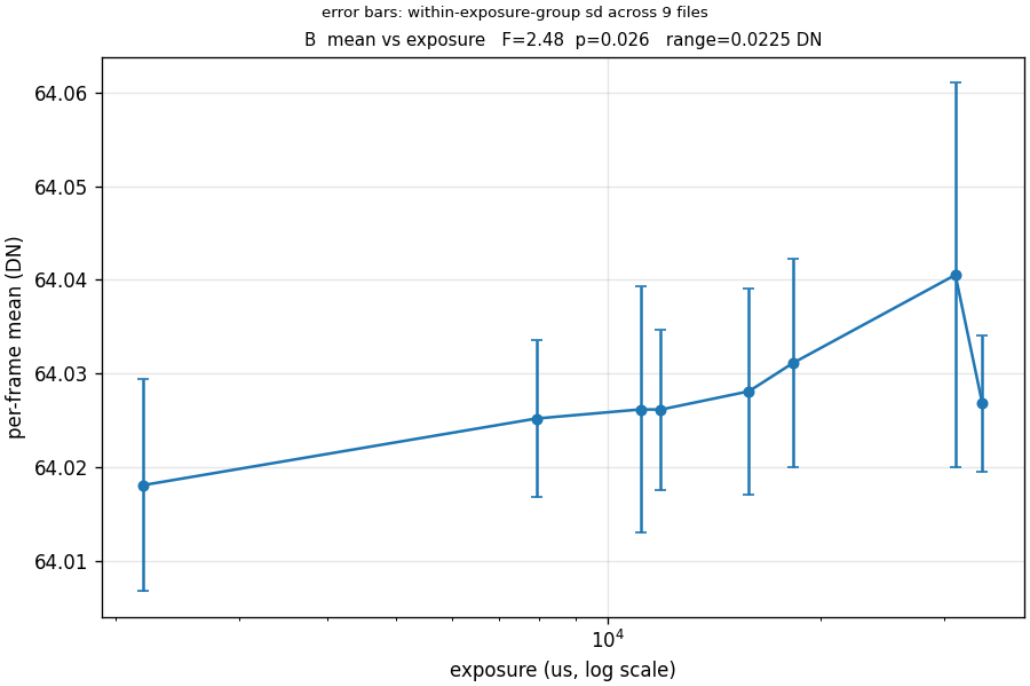


Figure 6: Per-frame mean dark level vs. exposure time, restricted to the 8 exposure times common to all nine archives. Error bars are the within-exposure-group standard deviation across archives.

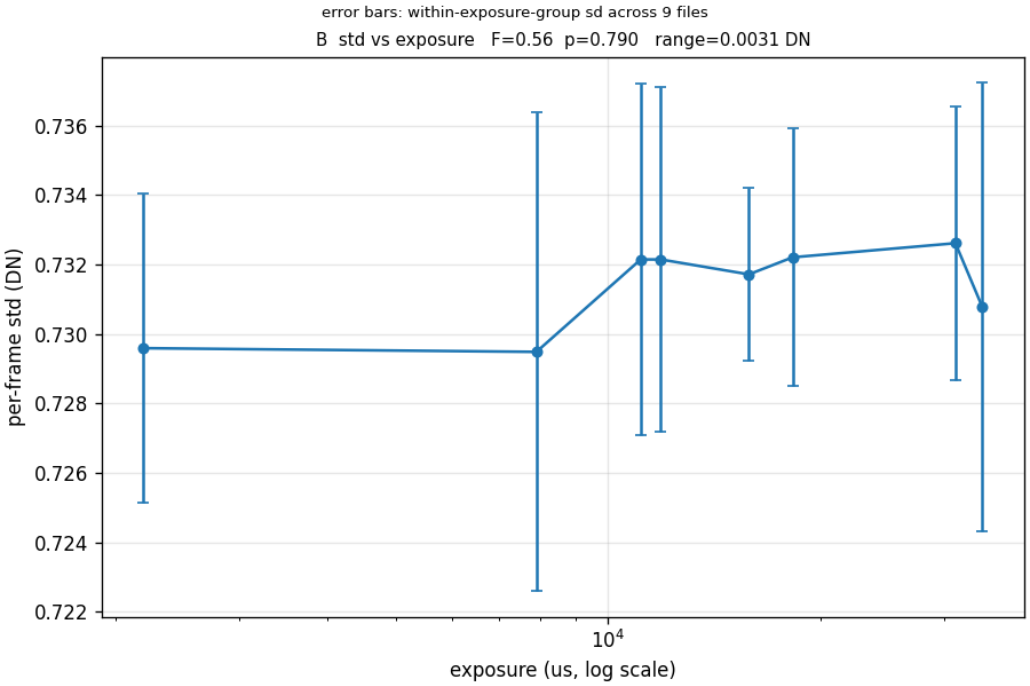


Figure 7: Per-frame standard deviation vs. exposure time, same restriction as Figure 6.

9 Combining it all: the best per-pixel dark estimator

Why we asked this. Everything above answers a piece of the puzzle. This section puts the pieces together into an actual answer to the practical question: given a new lit frame, what should be subtracted from it, pixel by pixel?

Restating the model with what we now know. Recall Equation 1, $d_i(k) = \mu(k) + p_i + \varepsilon_i(k)$. We now know: $\mu(k)$ can be measured directly and freshly for any new dark frame, with negligible sampling error of its own (it is already an average over all 23,790 pixels of one frame); its drift over the timescales we could test is not detectably different from zero (§7); it does not detectably depend on exposure duration by an amount that matters (§8). p_i is mostly indistinguishable from noise for a typical pixel, but real and reproducible for a small population of hot pixels (§6). $\varepsilon_i(k)$ is fresh, independent noise that cannot be predicted for any specific future frame, by construction.

An important, easy-to-miss point about what "best" means here. No procedure, however good, can predict $\varepsilon_i(k_{\text{new}})$ — the specific noise realization affecting a future lit frame's own dark contribution — because it is a fresh, independent draw that no measurement taken at another time could ever have observed. That is not a limitation of the estimator; it is a limitation of the physical situation. The best any estimator can do is hit the *predictable* part of the dark signal, $\mu(k_{\text{new}}) + p_i$ — its expected value averaged over that unavoidable fresh noise. Every comparison below therefore judges candidate estimators by how close they come to $\mu(k_{\text{new}}) + p_i$, not by pretending to explain away noise nothing could explain away.

Estimating p_i from history: the shrinkage estimator. p_i is never observed by itself, only mixed with fresh noise, $d_i(k) - \mu(k) = p_i + \varepsilon_i(k)$. Averaging this quantity over n historic frames for one pixel gives

$$r_i = \frac{1}{n}\sum_{k=1}^{n}\big(d_i(k) - \mu(k)\big) = p_i + \frac{1}{n}\sum_{k=1}^{n}\varepsilon_i(k),$$

an unbiased estimate of p_i whose own sampling variance is $\mathrm{Var}(r_i) = \sigma_\varepsilon^2/n$ (averaging n independent noise draws divides variance by n). If, in addition, the true pixel-to-pixel spread of p_i itself, σ_p^2, is known, there is a simple, elementary way to combine the two pieces of information we have about this one pixel's p_i: a working assumption that a typical pixel's offset is near zero, with "precision" (inverse variance) $1/\sigma_p^2$; and the specific measurement r_i for this pixel, with precision n/σ_ε^2. The minimum-variance way to combine two independent, unbiased pieces of information about the same unknown is to weight each by its own precision and renormalize. Doing that here gives

$$B = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_\varepsilon^2/n}, \qquad \hat{p}_i = B\, r_i, \qquad 0 \le B \le 1. \tag{2}$$

(Readers familiar with Bayesian statistics will recognize this as the posterior mean of the standard normal-normal conjugate update; it is derived here from ordinary variance arithmetic because nothing more is needed.) B close to 1 means a pixel's own history is trustworthy and a fresh single dark frame would add little beyond it; B close to 0 means the apparent pixel-to-pixel spread is mostly sampling noise, and the population average — effectively, ignoring p_i altogether — is the better bet.

Estimating the two variances from the data, not assuming them. σ_ε^2 is the leftover spread of $\varepsilon_i(k)$ once both $\mu(k)$ and r_i are removed, averaged over pixels: $\sigma_\varepsilon^2 = \frac{1}{N_{\text{px}}}\sum_i \mathrm{Var}_k\big(d_i(k) - \mu(k) - r_i\big)$; this should sit close to the single-frame standard

deviation already seen in Figure 1. σ_p^2 is trickier, because the raw spread of r_i *overstates* it: $\mathrm{Var}(r_i)$ mixes the real between-pixel spread with r_i's own sampling noise, $\sigma_p^2 + \sigma_\varepsilon^2/n$ (this is the same "total variance splits into a real part and a sampling-noise part" idea used throughout, and is the same decomposition a random-effects ANOVA performs, with pixel playing the role of "group"; B above is that decomposition's intraclass correlation). Removing the known sampling-noise contribution gives the corrected estimate

$$\sigma_p^2 = \max\big(\mathrm{Var}(r_i) - \sigma_\varepsilon^2/n,\ 0\big),$$

clipped at zero because a negative raw result simply means the apparent pixel-to-pixel spread is no larger than sampling noise predicts on its own — there is no fixed pattern left to estimate. Both quantities, together with B, are computed by `estimate_dark_offset(samps, ch)`. On the nine-archive pool: $\sigma_\varepsilon = 0.731\,\mathrm{DN}$, $\sigma_p = 0.028\,\mathrm{DN}$, $B = 0.131$ at $n = 100$.

Four candidate estimators, compared. With μ_{new} standing for a freshly measured dark frame's own spatial mean, we compare four ways of predicting $\mu_{\mathrm{new}} + p_i$ for a future lit frame:

A — frame mean only
$\hat{d}_i = \mu_{\mathrm{new}}$, applied uniformly to every pixel (i.e. assume $p_i = 0$). Its error against the target is simply p_i itself, so its root-mean-square error over all pixels is σ_p.

B — raw pooled history
$\hat{d}_i = \mu_{\mathrm{new}} + r_i$, using the plain historic average with no shrinkage. Its error is $-\frac{1}{n}\sum_k \varepsilon_i(k)$, RMS $= \sigma_\varepsilon/\sqrt{n}$.

C — shrunk pooled history
$\hat{d}_i = \mu_{\mathrm{new}} + Br_i$, Equation 2. A short calculation (expand $p_i - Br_i = (1-B)p_i - B\cdot\frac{1}{n}\sum_k \varepsilon_i(k)$, take the variance of both independent terms, and substitute B's definition) gives mean-squared error exactly $(1-B)\,\sigma_p^2$, so RMS $= \sqrt{(1-B)}\,\sigma_p$ — always at least as good as A, by construction.

D — today's practice
$\hat{d}_i = d_i(k_{\mathrm{match}})$: subtract one specific historic dark frame's own raw per-pixel value, implicitly treating its $\mu(k_{\mathrm{match}})$ as a stand-in for μ_{new}. Its error against the target is $\varepsilon_i(k_{\mathrm{match}})$ (plus any real difference between $\mu(k_{\mathrm{match}})$ and μ_{new}, which §7 found not to be detectably different from zero), so RMS $= \sigma_\varepsilon$.

Table 1: The four estimators, evaluated with the numbers measured on this sample ($\sigma_\varepsilon = 0.731\,\mathrm{DN}$, $\sigma_p = 0.028\,\mathrm{DN}$, $n = 100$, $B = 0.131$).

Estimator	What it subtracts	RMS error formula	RMS error (DN)
A — frame mean	μ_{new}	σ_p	0.028
B — raw pooled history	$\mu_{\mathrm{new}} + r_i$	$\sigma_\varepsilon/\sqrt{n}$	0.073
C — shrunk pooled history	$\mu_{\mathrm{new}} + Br_i$	$\sqrt{1-B}\,\sigma_p$	0.027
D — today's practice	$d_i(k_{\mathrm{match}})$	σ_ε	0.731

What this means. Two results stand out. First, C improves on A by only about 7% (0.027 vs. 0.028 DN) — exactly what a small B predicts, and confirmation that chasing a fully pooled per-pixel history is not worth the added bookkeeping for an ordinary pixel. Second, and by far the larger finding: A beats D, today's actual practice, by a factor of about 26 in RMS error (0.028 vs. 0.731 DN). Subtracting one matched historic dark frame's own raw pixel values — which feels

like the more careful, "use the real measurement" choice — is in fact dramatically worse than simply subtracting that frame's spatial average uniformly, because it needlessly imports that one frame's entire share of fresh, unpredictable noise, σ_ε, in place of the far smaller, genuinely predictable quantity, σ_p. (For completeness: A would lose to B, the raw pooled history, only once n exceeds $\sigma_\varepsilon^2/\sigma_p^2 \approx 662$ historic frames — far more than are presently collected for any one pixel.)

The exception: hot pixels. This ranking is for a *typical* pixel, where p_i is small and mostly indistinguishable from noise. §6 identified a small, reproducible population where that is not true. `estimate_dark_offset` flags these automatically: a pixel is marked an outlier when its historic residual r_i exceeds five times r_i's own predictive spread, $\sqrt{\sigma_p^2 + \sigma_\varepsilon^2/n}$ — the full uncertainty an ordinary pixel's r_i can carry, sampling noise included, rather than σ_p alone. (An earlier version of this check compared against σ_p alone and flagged 1,589 of 23,790 pixels — clearly wrong, since it was penalizing ordinary pixels for r_i's own sampling noise, not for anything peculiar to the pixel. Correcting the comparison brought the count down to a small, plausible number.) On this sample, 12 pixels out of 23,790 are flagged this way, with offsets from 0.4 to 1.4 DN — large enough, and reproducible enough (§6), that estimator A's flat assumption $p_i = 0$ leaves a real, avoidable bias at exactly these locations. For this small list, estimator C (or D) is the better choice; for everywhere else, A is.

10 Decisions

Two concrete changes follow from the six tests above. Both are proposed, not yet implemented, in `docs/design-updates-prompt-1.md`, which is the place to look for the implementation plan, cost, and open questions; this section states only the reasoning, which is what this document is for.

Decision 1: subtract the fresh dark frame's spatial mean, not its raw per-pixel array

Everywhere except the identified hot pixels, §9 shows that subtracting a freshly measured dark frame's own spatial average, uniformly across every pixel, is about 26 times more accurate (in RMS error against the predictable part of the dark signal) than the present practice of subtracting that one frame's raw pixel-by-pixel values. This is because a frame mean, averaged over 23,790 pixels, already carries almost none of a single dark frame's own read noise, while a raw per-pixel value carries all of it. For the small, explicitly identified set of hot pixels, their own pooled historic correction (or a matched raw frame) should still be used, since for them the fixed offset p_i is real, sizeable, and well estimated by history.

Decision 2: replace the exposure ladder with a before/after pair

§8 found no dark-level or noise dependence on exposure duration large enough to matter operationally: the detectable effect on level is about 30 times smaller than a single frame's own noise. This means a full ladder of dark exposures, one per exposure time used in the lit sweep, is not needed for accuracy. What *does* matter, per Decision 1, is having a good, fresh estimate of $\mu(k)$ close in time to the lit frames it will correct — and §7 could not test the one timescale that matters most for that (dark-to-lit, within one sweep), because every archive in this sample keeps only a single ("after") dark batch. Reinstating a "before" batch as well as an "after" batch, bracketing both the pre-image sweep and the measurement sweep, closes that gap directly: it gives two independent, closely-timed estimates of $\mu(k)$ to average or compare, and, as a side benefit, makes the previously untestable middle timescale testable for the first time.

11 Provenance and open items

- Code: `desktop/padanalysis/analysis_2.py`, functions `dark_analysis1`, `dark_analysis2`, `dark_cdf_check`, `dark_stability_check`, `dark_temporal_check`, `dark_exposure_check`, `estimate_dark_offset`, `predict_dark`.
- Sample set: `C1017`–`C1025`, measurement sweep, channel `B`, 100 pooled dark frames.
- Related documents: `docs/design-updates-prompt-1.md` (the implementation plan for both decisions above); `docs/reference_framework_preimage_and_chips.md` (the earlier, pre-2026-08 finding that dark-frame spatial structure is negligible, on archives that did keep before/after pairs); `tex/R_inf_guide.tex` (where `sig = lit - dark` enters the larger photometric chain).
- Open items, not yet acted on: everything above is channel `B` only — `R`, `Gr`, and `Gb` have not been run through this same battery of tests; the apparent session-boundary step pattern in Figure 5 is unconfirmed, with only three groupings observed; and both decisions in §10 remain proposals until implemented and re-validated on data captured under the new protocol.